THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE TWO-CHAPTER TEST SET B

Chapters 9 & 10: **The Amazing World of Solutes, Solvents and Solutions, Light: Mirrors and Lenses** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The density of water is about: 1

(a) 10 g/cm³ **(b)** 1 g/cm³ **(c)** 0.1 g/cm³ **(d)** 100 g/cm³

2. When ice forms, water particles: 1

(a) come closer **(b)** disappear
(c) arrange to take up more space **(d)** turn to gas

3. An egg sinks in tap water but floats in salt water because salt water has: 1

(a) greater density **(b)** less density **(c)** no density **(d)** less mass

4. A gas–liquid solution is: 1

(a) soda water **(b)** brass **(c)** air **(d)** sugar syrup

5. A solution can be turned from saturated to unsaturated at the same temperature by: 1

(a) shaking it **(b)** adding more solute
(c) cooling it **(d)** adding more solvent

6. When an object is far from a concave mirror, the image is: 1

(a) invisible **(b)** erect **(c)** always enlarged **(d)** inverted

7. A convex lens makes light rays: 1

(a) diverge **(b)** stop **(c)** stay parallel **(d)** converge

8. The bending of light when it passes from one transparent medium to another is: 1

(a) diffraction **(b)** absorption **(c)** reflection **(d)** refraction

9. Which pair is correctly matched? 1

(a) Concave mirror – converging **(b)** Plane mirror – converging
(c) Convex mirror – converging **(d)** Concave lens – converging

10. Rough surfaces: 1

(a) do not reflect at all **(b)** only absorb
(c) only refract **(d)** also reflect light (diffuse reflection)

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. A cube has mass 54 g and volume 20 cm³. Find the density. 2

12. Find the volume of 500 g of a substance with density 2 g/cm³. 2

13. What is refraction of light? Give one example. 2

14. Why does a pencil in water appear bent? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. Find the mass of 2 litres of water (density 1 g/cm³; 1 L = 1000 cm³). 3

16. Explain why light reflects in many directions from a rough wall. 3

17. Explain why oil floats on water. 3

OR

State the effect of stirring on dissolving. Does it change solubility? 3

18. Describe how you would identify a mirror by looking at your face in it from near. 3

OR

Why are ENT specialists' mirrors concave? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–100 words each.

19. Why do some objects float and others sink? Explain with density and give examples of wood, iron and ice. 5

OR

Explain saturated solution preparation and how to convert it into an unsaturated solution by two methods. 5

20. A ray makes 30° with the normal on a mirror. 5
(i) Find angles of incidence and reflection.
(ii) Angle between the ray and mirror.
(iii) Angle between incident and reflected ray.
(iv) What if the mirror is tilted to make the ray fall along normal?

OR

Explain how a solar cooker works and why concave mirrors are suitable. 5

END OF QUESTION PAPER — **Total: 40 Marks**

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